1. About this guide

> ⚠️ **Caution** This manual explains how to operate DWOS Web safely and effectively. Read it before designing your first restoration. Where the application itself shows a warning (yellow ⚠ banners, red status text, confirmation dialogs), the warning in the running software always takes precedence over anything written here.

This guide applies to **DWOS Web**, the browser-based re-implementation of the DWOS restoration CAD/CAM workflow. It covers the CAD workstation reachable from the `/cad` route, the import of surface scans, AI-assisted tooth segmentation, and the design and export of restorative prosthetics.

The name *DWOS Web* in this guide refers to this single web application. It is a companion to the coDiagnostiX Web implant-planning clone that lives in the same repository; the two share a data directory and several services, and a separate manual documents coDiagnostiX Web.

The manual is delivered in electronic form:

- as Markdown files under `docs/manual-dwos/` in the repository, and
- rendered to a printable PDF (`docs/manual/DWOS-Web-Manual.pdf`).

You can print any chapter with your browser's print function (Ctrl+P).

1.1 Disclaimer

DWOS Web is a **demonstration re-implementation** built for training, evaluation and software-development purposes. It is **not a certified medical device**, has not undergone regulatory clearance (no CE mark, no FDA clearance), and must not be used to design restorations for the treatment of real patients.

Beyond that fundamental restriction, the same professional logic applies that governs any restoration-design software:

- The operator is responsible for verifying every result the software computes — automatic tooth segmentation, margin detection, proposed anatomy and intrados surfaces are *aids*, never decisions.
- The operator is responsible for the correctness, completeness and adequacy of all data entered: orders, imported surface scans, restoration parameters and material selections.
- Design quality depends on the quality of the input scans. A clean prepared stump with a clearly captured margin is required for every restoration; low-resolution, incomplete or noisy scans degrade every downstream step.

1.2 License, trademarks and other rights

The application is built on open-source components, each under its own license. Its restoration design is performed by a vendored build of the **Chili3D** CAD engine (`vendor/chili3d`), and the surrounding web application uses SvelteKit and Svelte (MIT), Bun (MIT) and three.js (MIT). The complete list with version numbers is the authoritative source for third-party attribution in the running build.

All example data shipped with the application is generated programmatically or anonymized and contains no real patient information.

2. Introduction and overview

2.1 Intended use

DWOS Web reproduces the workflow of dental restoration CAD/CAM software in a web browser: it imports 3D surface scans of a prepared clinical situation, lets the operator design dental restorative prosthetics on those surfaces, and exports the design as an open mesh for manufacturing.

Because this build is a demonstration re-implementation (see chapter 1.1), its intended use is **training, evaluation and development** — not the design of restorations for patient treatment.

2.2 Device description and features

DWOS Web imports and visualizes 3D scan data of the clinical situation and lets the operator design restorations on it inside an embedded CAD workstation. The design result is exported as an open mesh for automated manufacturing.

The restoration types covered by the original DWOS suite are crowns, bridges, abutments, screw-retained bars and bridges, partial frameworks, full dentures, bite splints and models. In this demonstration build the in-scope and planned scope are:

Restoration type	Status in this demo
Single full-contour crown	In scope (worked example, chapter 5)
Bridges	Planned
Abutments	Planned
Bars and partial frameworks	Planned
Full dentures	Planned
Bite splints	Planned
Models	Planned

A real, working feature already built into DWOS Web is **AI tooth segmentation of intraoral scans**: an imported intraoral scan is sent to a vendor segmentation model that returns per-vertex tooth labels, which are mapped to **FDI** tooth numbers. This auto-tagging gives each tooth in the scan its dental notation and drives the worked crown example.

Device variants and configuration

The original DWOS suite ships in three variants — **DWOS** (full functionality for dental laboratories), **DWOS Easy** (a.k.a. Straumann Nova™, a simplified subset for laboratories) and **DWOS Chairside** (a simplified subset for dental clinics). DWOS Web is a single web application; it does not reproduce the variant split. Where this manual refers to a variant it is describing the original product for orientation only.

Principles of operation

DWOS Web provides solutions for computer-aided design (CAD) of dental restorations. It is an open system: it operates on 3D surface scans supplied in open file formats and produces design results as open meshes (STL) for any compatible manufacturing chain. All 3D data is interpreted in millimetres.

The software supports digital realization of conventional dental restoration design; it does not introduce novel clinical features relative to conventional dentistry.

2.3 Indications

Within its demonstration scope, the software supports the design of dental restorative prosthetics by dental professionals who have appropriate knowledge in the respective field of application: importing and visualizing 3D scan geometry, labelling teeth, and designing restorations on the prepared situation for export to a manufacturing chain.

2.4 Accessories and products used in combination

Input — scanners. DWOS Web operates on 3D surface scans from intraoral or desktop scanners. Any scanner that exports 3D data in an open file format (**STL**, **PLY** or **OBJ**) and is certified for dental scanning may, in principle, be suitable; the operator is responsible for ensuring the files meet the data input requirements below. Intraoral scans for AI auto-tagging are processed as binary glTF (GLB) per-vertex meshes.

Output — manufacturing. Design results are exported as open **STL**. Any 3D manufacturing chain (milling or printing) that accepts STL and is certified for dental restoration fabrication may, in principle, fabricate them; the operator is responsible for ensuring the results meet the requirements for the dental restoration.

Software used in combination. DWOS Web shares a repository and data directory with the **coDiagnostiX Web** implant-planning clone. In the original products the two exchange cases through DWOS Synergy; this demonstration build documents the interplay but does not reproduce the proprietary Synergy transport.

> ⚠️ **Caution** The software performance depends on the quality and accuracy of the imported 3D scans. The production of scans, and of the manufactured restoration, lies fully within the responsibility of the user. A clean prepared stump with its margin clearly captured is required for every restoration.

2.5 Contraindications

Do not use the software:

- for designing restorations for the treatment of real patients (demonstration build — see chapter 1.1),
- in direct contact with a patient, or in combination with life-sustaining devices,
- with scans that were imported despite warnings you do not fully understand, or whose margin cannot be identified with confidence.

2.6 Precautions

- **Verify every automatic result.** AI tooth segmentation, margin detection and proposed anatomy are aids; check each before relying on it.

- **Verify scan quality before design.** A noisy, incomplete or low-resolution scan produces an unreliable restoration regardless of how the design tools behave.
- **Use a strong password** for any deployed instance to reduce the risk of intrusion.
- **Verify dimensions.** All geometry is interpreted in millimetres; confirm exported meshes are at the expected scale before manufacturing.

2.7 Compatibility information

The client requires a desktop browser with **WebGL 2**; the embedded CAD workstation renders in WebGL. Tested with current Chromium-based browsers; equivalent-generation Firefox and Safari releases are expected to work. The server requires the **Bun** runtime. Supported import formats are STL, PLY and OBJ; export is STL.

2.8 Data protection

- Access is restricted by user accounts shared with the coDiagnostiX Web application (password hashing, optional two-factor authentication, login-attempt lockout).
- Orders, scans and design files live in the server data directory; deleting a record removes its associated files, not only the database rows.
- The optional AI-segmentation backend transmits scan geometry to an external model endpoint; see chapter 3 for how to configure or disable it. Do not send identifiable data to an external endpoint you do not control.

2.9 Data backup

Everything DWOS Web stores lives in the server data directory: the database plus all imported scans and exported designs. Back the directory up as a whole, regularly and before every update. See chapter 6.

2.10 Disposal (data deletion)

Electronic data takes the place of a physical device here: when an order or design is no longer needed, export the design if retention is required, then delete the record. Deletion removes all files belonging to the record from the data directory.

3. Installation (deployment)

DWOS Web is a web application: there is nothing to install on the workstation — operators only need the URL and an account. "Installation" here means deploying and serving the web app and, optionally, configuring the AI-segmentation backend.

3.1 Building and serving the application

DWOS Web is served from the same SvelteKit + Bun project as coDiagnostiX Web; the CAD workstation reachable at `/cad` is built together with the rest of the application.

```
bun install
bun run build                 # also builds the embedded Chili3D CAD workstation
PORT=3000 ORIGIN=https://your-host bun run build/index.js
```

`ORIGIN` must be set to the public URL the browser uses (scheme + host + port) so that form actions and the CAD `postMessage` bridge accept requests; a mismatch blocks those interactions. Place the server behind HTTPS for any non-local use.

The repository is self-contained — the embedded CAD's complete source ships under `vendor/chili3d`, so deployment needs no access to external code hosting.

> 💡 **Hint** For evaluation, `bun run dev` starts a development server (default port 5173) with the same feature set. Open `/cad` to reach the restoration workstation directly.

3.2 Data directory

The database and all order files (imported scans, exported designs) live in the server data directory (`CDX_DATA_DIR`, default `./data`). Back it up as a whole; the directory is portable and may be moved to a new server to preserve every order and design.

3.3 Browser requirements

A desktop browser with **WebGL 2** is required — the CAD workstation will not render otherwise. A display of 1600 × 900 or larger and a mouse with a wheel and middle button are recommended for comfortable navigation. See chapter 8 for full requirements.

3.4 First run and login

On first start, create the initial account (or sign in with credentials provisioned by an administrator). Accounts and authentication are shared with the coDiagnostiX Web application in the same deployment. After signing in, open the **`/cad`** route to load the restoration workstation.

3.5 Optional AI-segmentation credentials

AI tooth segmentation of intraoral scans calls an external vendor model. Configure it through the server environment before starting the server:

Variable	Purpose	Default
`CDX_AISEG_URL`	Base URL of the segmentation backend	`https://pbapi.becertain.ai`
`CDX_AISEG_EMAIL`	Account e-mail for the backend	_(unset)_
`CDX_AISEG_PASSWORD`	Account password for the backend	_(unset)_

The backend is read from the server process, so set these before launching the server. When credentials are not configured, AI auto-tagging is unavailable; scans can still be imported and labelled manually.

> ⚠️ **Caution** The segmentation backend receives the scan geometry you submit. Do not enable it for data you are not permitted to send to an external endpoint.

4. Basic principles and user interface

4.1 Getting acquainted

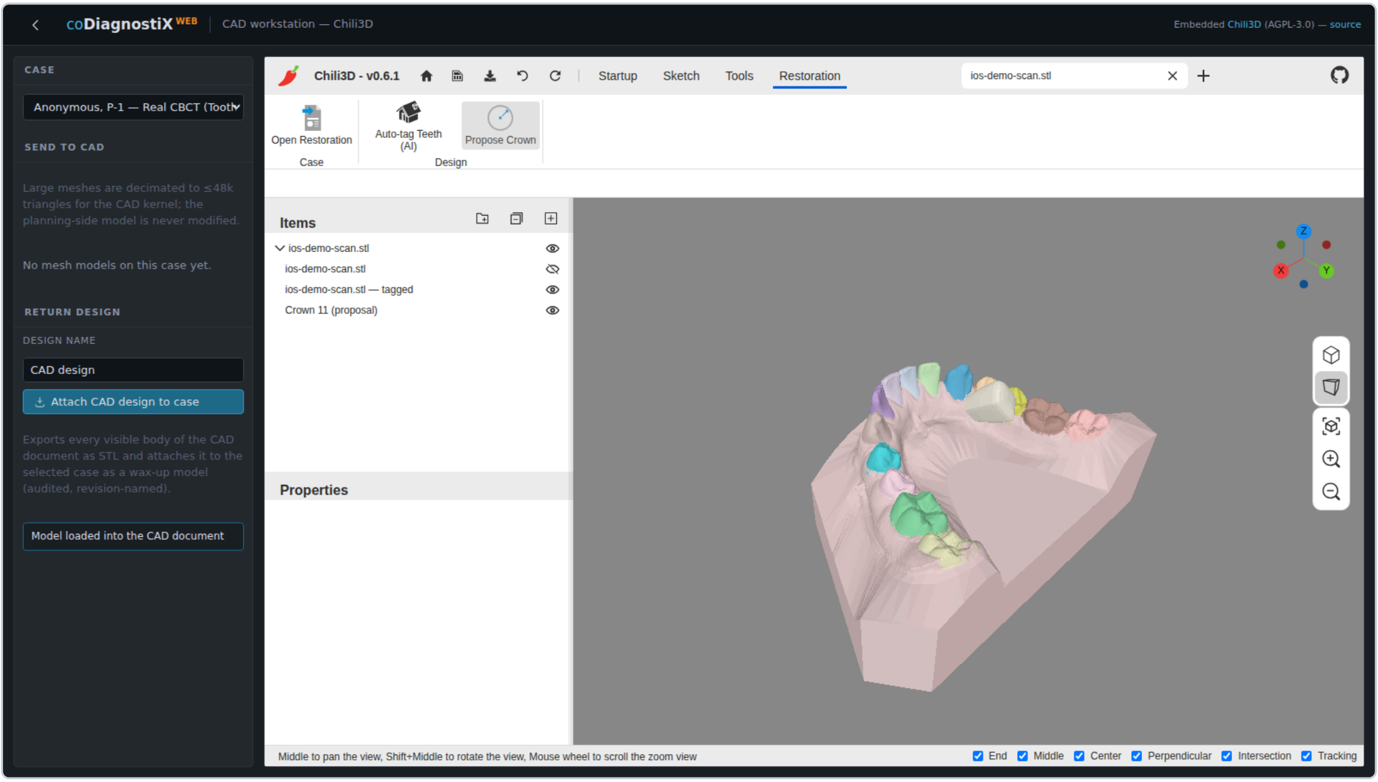
DWOS Web runs entirely in a web browser. The restoration workstation is an embedded CAD application (a vendored build of the open-source Chili3D engine) extended with a **Restoration** workflow for dental design. No desktop installation is required; the workstation is reached from the web app at the `/cad` route.

> _DWOS Web is a demonstration / development build and **not a medical device**. It is intended to illustrate the restoration workflow, not for clinical production._

4.2 Starting the application

1. Sign in to DWOS Web (see *Installation*, ch. 3).
2. Open a case. The embedded CAD workstation loads; when the status panel reads **CAD ready**, the engine is initialised and ready for input.

The workstation is organised as a left-hand **Case** panel (case selection, model import, design return), a central 3-D viewport, an **Items** tree and **Properties** panel, and the command **ribbon** along the top.



4.3 The Restoration ribbon and stations

The ribbon carries the general CAD tabs (**Startup**, **Sketch**, **Tools**) plus a dedicated **Restoration** tab that drives the dental workflow. Its commands are grouped into stations that follow the design order, left to right:

- **Case** — *Open Restoration*: brings the restoration workflow online for the active document.

- **Design** — *Auto-tag Teeth (AI)* (identify each tooth by FDI from the scan); *Select Tooth* (click a tooth to choose the working site); *Margin Line* (trace the selected tooth's emergence line on the surface); *Insertion Axis* (show its insertion/withdrawal direction); *Propose Crown* (place a library tooth at a detected edentulous site).
- **Output** — the design is returned to the case as an STL wax-up from the **Case** panel (*Attach CAD design to case*).

4.4 Mouse and navigation

The viewport uses the engine's standard navigation:

Action	Result
Left click	Select / pick a point
Right click	Context menu / validate
Mouse wheel	Zoom in / out
Press and hold the **middle** button + drag	Pan the view
Shift + middle button + drag	Rotate the view
Double-click left	Centre and fit the view

The navigation cube at the top-right snaps the camera to standard views.

4.5 Restoration design overview

The restoration engine works directly on the **surface-scan mesh** of the clinical situation (intraoral or model scan). A design is produced in three moves:

1. **Recognise** — an AI model segments the scan into individual teeth (by FDI number) and gingiva, and flags any edentulous gap.
2. **Propose** — a library tooth is placed at the restoration site, oriented from the scan itself (the occlusal axis is derived from the gingiva-to-teeth direction, the mesiodistal axis from the neighbouring teeth), so no manual alignment is required.
3. **Output** — the proposal is a normal mesh body and is exported as open **STL** for downstream manufacturing.

The step-by-step instructions in chapter 5 walk through one complete case.

5. Step-by-step instructions

This chapter walks through one example restoration case in DWOS Web: a **single full-contour crown** for a missing upper central incisor (FDI 11), designed from an intraoral surface scan. The figures are captured from the running application.

> _Example only. DWOS Web is a demonstration build and not a medical device; do not use its output for clinical treatment._

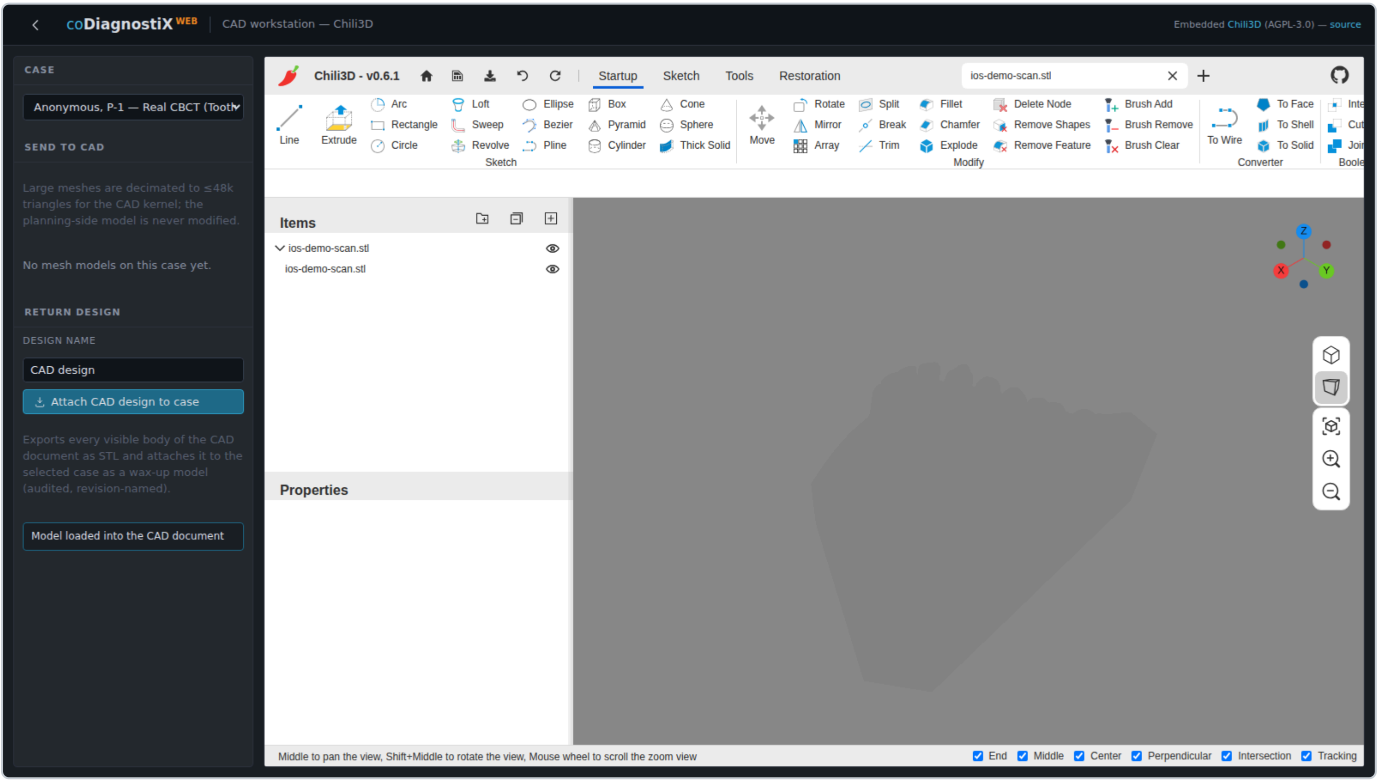
5.1 Order creation

Open a case and start the workflow:

1. Sign in and open the case for the patient.
2. The CAD workstation loads; wait for **CAD ready**.
3. On the **Restoration** ribbon tab, in the **Case** group, click **Open Restoration**.

5.2 Scan import

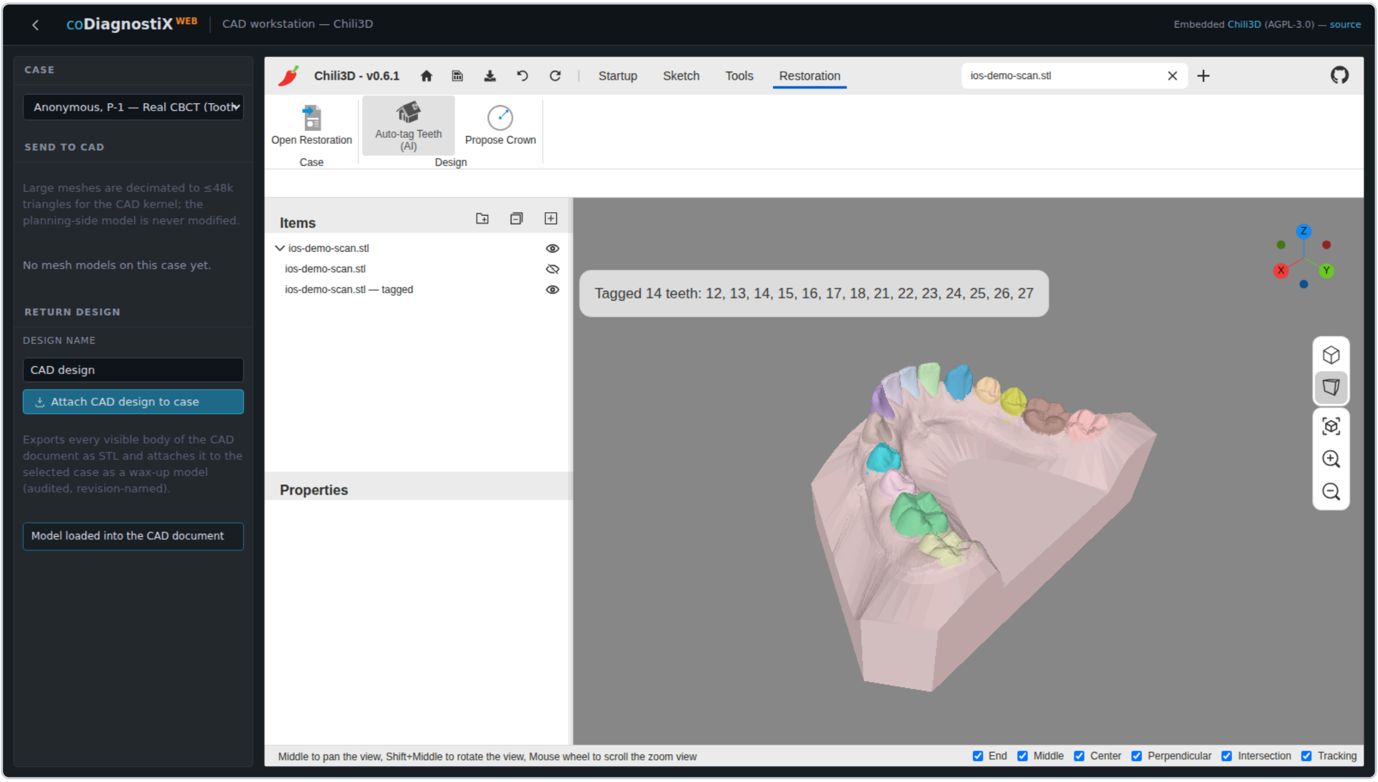
The design is built on a surface scan of the clinical situation. Load the scan into the document as a mesh (supported formats: **STL**, **PLY**, **OBJ**). The scan appears in the **Items** tree and in the viewport.



5.3 Auto-tagging (AI tooth segmentation)

In the **Design** group, click **Auto-tag Teeth (AI)**.

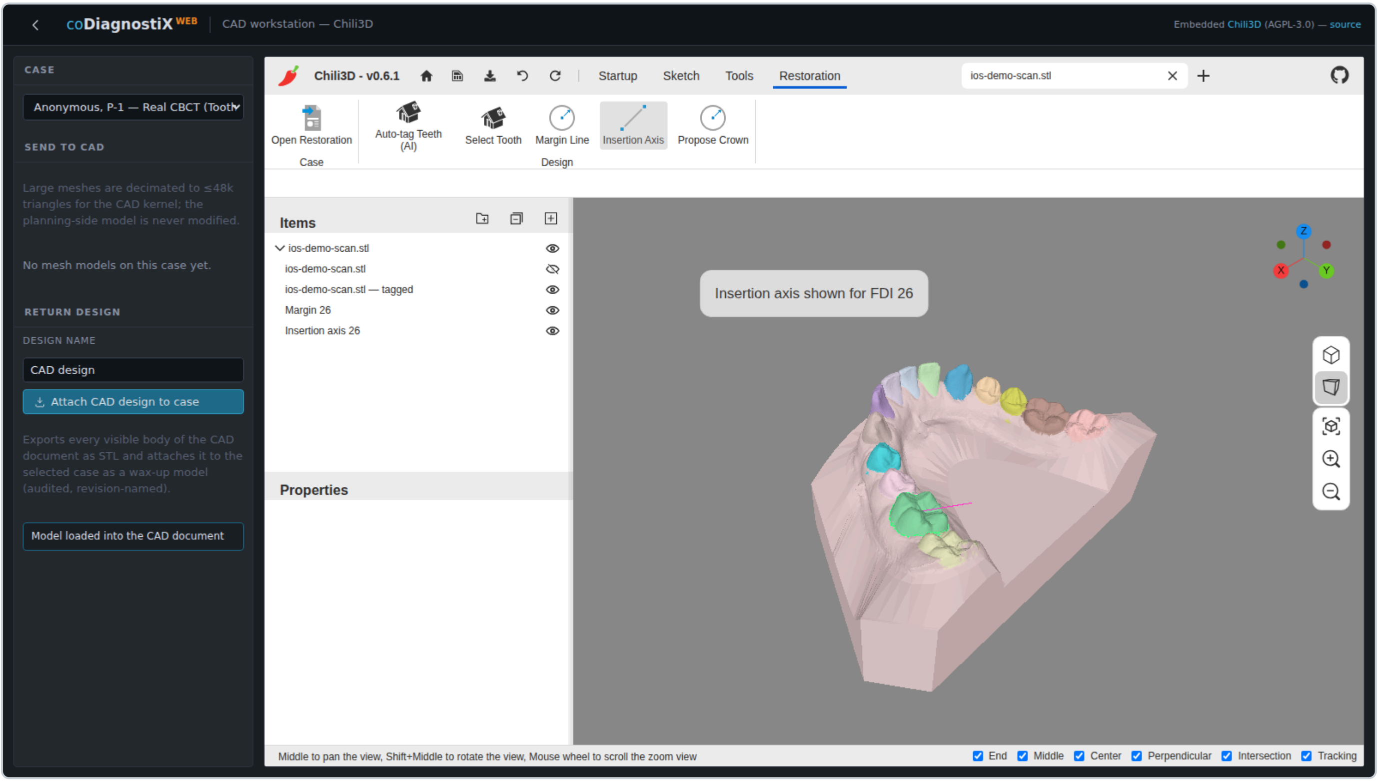
The scan is sent to the tooth-segmentation model. Within a few seconds the result is returned and the scan is rebuilt as a per-tooth coloured mesh: every tooth is identified by its **FDI number**, the gingiva is separated, and any edentulous gap is detected. A confirmation reports the teeth found — in this example **14 teeth (FDI 12–18, 21–27)**, with the central incisor **11** absent.



5.4 Select tooth, margin and insertion axis

First choose the working tooth, then trace its margin and show its insertion axis — all driven by the segmentation:

1. **Design ▸ Select Tooth**, then click a tooth on the scan. Its FDI number is reported (picking resolves the exact triangle and reads its tooth label).
2. **Design ▸ Margin Line** traces the selected tooth's **emergence line** — the boundary between the tooth and the gingiva — directly on the scanned surface (a curve riding the mesh), and adds it as a *Margin* overlay.
3. **Design ▸ Insertion Axis** shows the tooth's insertion/withdrawal direction (the occlusal axis, derived from the gingiva-to-teeth direction) as an *Insertion axis* overlay.

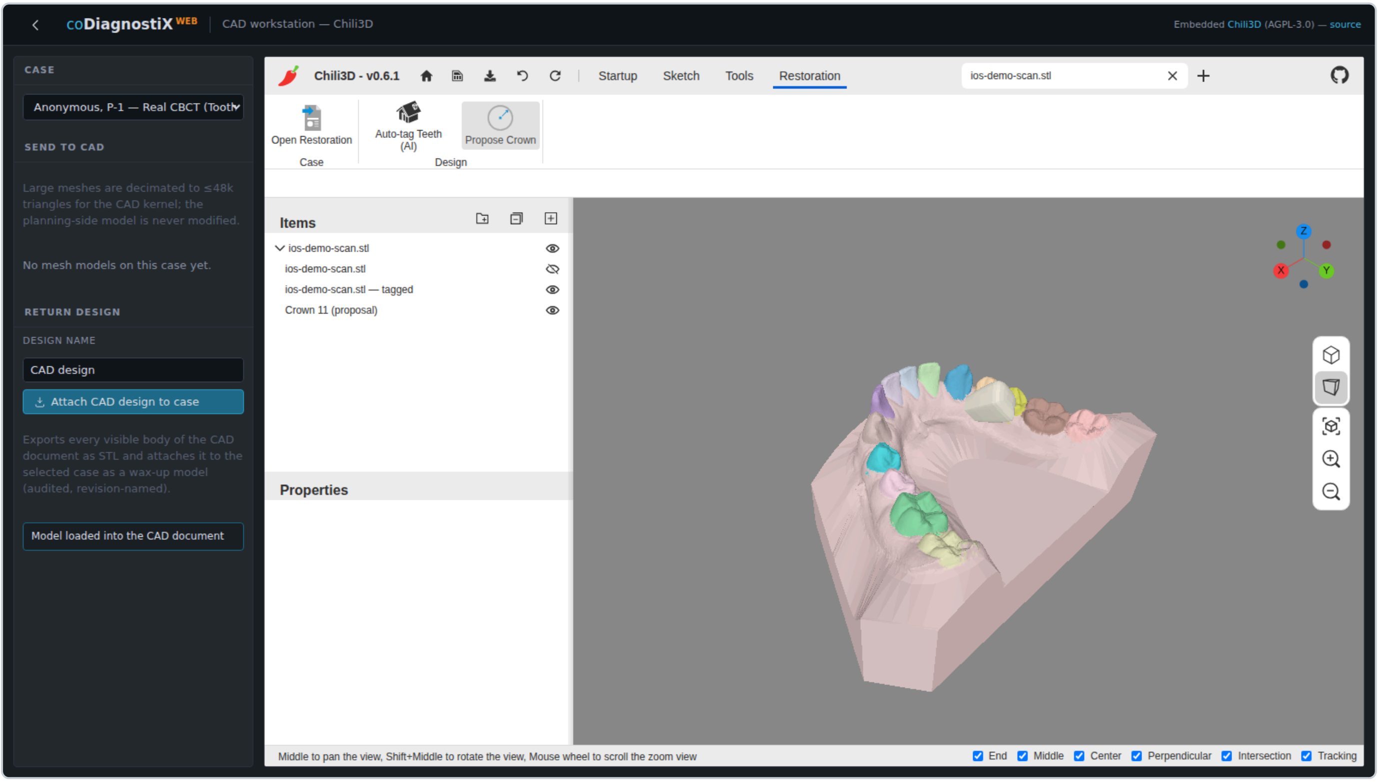


> The cement-gap intrados that offsets the fitting surface inside this margin is part of the planned design refinement (see 5.6).

5.5 Anatomy — crown proposal

In the **Design** group, click **Propose Crown**.

The application reads the segmentation, identifies the single-tooth gap and its neighbours (here **FDI 11**, between **12** and **21**), and places a library tooth at that site. The proposal is oriented automatically from the scan — the occlusal axis from the gingiva-to-teeth direction and the mesiodistal axis from the neighbouring teeth — so no manual alignment is needed. The crown is added to the **Items** tree as *Crown 11 (proposal)*.



5.6 Intrados

The cement-gap **intrados** (the fitting surface offset from the preparation, bounded by the margin) is part of the planned design refinement (chapter 2 roadmap). The full-contour proposal from 5.5 is exported as-is in this demonstration build.

5.7 Export

Return the design to the case as manufacturing output:

1. In the **Case** panel, enter a **design name**.
2. Click **Attach CAD design to case**.

Every visible body in the document is exported as **STL** and attached to the case as a revision-named wax-up model, ready to be sent for manufacturing.

6. Maintenance

The application itself needs no scheduled maintenance — there are no dongles, local installations or wear parts. Keeping a deployment healthy means three things:

Back up the data. Everything lives in the data directory (`CDX_DATA_DIR`): the database plus all imported scans and exported designs. Back the directory up as a whole, regularly and before every update. Individual designs can additionally be exported as STL for offline retention.

Update the software. Updates are deployed server-side (`git pull`, `bun install`, `bun run build`, restart); any database migrations run automatically on startup. The embedded CAD workstation is rebuilt as part of `bun run build`. Operators simply reload the browser. Review the changelog before updating a production deployment, and verify after the update that a representative order opens, a scan imports, the CAD workstation renders and an STL exports.

Watch the access and audit records. Sign-ins and integrity-relevant events are recorded together with the coDiagnostiX Web application sharing the deployment. Unexpected entries are the earliest indicator of misuse or misconfiguration.

> ⚠️ **Caution** Restoring a backup replaces the entire data directory — designs changed after the backup point are lost. Export current work before restoring.

7. Distributors and service

DWOS Web is a demonstration re-implementation and has no distributor network. For questions, defects and feature requests:

- **The manual sources** — `docs/manual-dwos/` in the repository; corrections are welcome as merge requests.
- **Issue tracker** — file reproducible reports (screenshots, the browser and version, and the steps to reproduce) with the repository's issue tracker.
- **Version identification** — quote the application version and the browser used in every report.

For the trademark owner's original DWOS product and its certified support and distribution channels, see chapter 1.2 — this project is not affiliated with them.

8. Technical data and label

8.1 Hardware and software requirements

Workstation (client):

Item	Minimum
Browser	Current Chromium-based browser (or an equivalent-generation Firefox/Safari) with **WebGL 2** — the CAD workstation will not render otherwise.
Display	1600 × 900 or larger; the design workstation uses a multi-panel layout.
Input	Mouse with wheel and middle button recommended (orbit, pan, zoom in the 3D view).
Memory	8 GB system RAM; the 3D view holds the active scan and design in GPU memory.

Server:

Item	Minimum
Runtime	Bun (bundles the SQLite driver); Linux x64 tested.
CPU / RAM	4 cores, 8 GB.
Deployment	`bun run build` + an `ORIGIN`-configured production server (chapter 3); place behind HTTPS for any non-local use.

8.2 Supported file formats

Direction	Formats
Scan import	**STL**, **PLY**, **OBJ** (surface meshes)
AI segmentation transport	binary glTF (**GLB**), per-vertex labelled
Design export	**STL** (open mesh for manufacturing)

All 3D data is interpreted in **millimetres**, even where the underlying exchange format carries no explicit dimension information.

8.3 Ports and services

Component	Default
Application server (production)	Port from `PORT` (example: 3000)
Development server	Port 5173 (`bun run dev`)
AI-segmentation backend	`CDX_AISEG_URL` (default `https://pbapi.becertain.ai`)
Data directory	`CDX_DATA_DIR` (default `./data`)

8.4 Identification (label)

The product identification of this build — name and version — is shown in the application. Quote the version shown there in every support request. This build repeats the demonstration-use disclaimer (chapter 1.1); it carries no medical-device label, UDI or manufacturer marking, because it is not a medical device.

9. Explanation of symbols

Symbols and notices used throughout DWOS Web and in this manual:

Symbol	Meaning
⚠ (yellow)	Advisory warning — scan quality, design rules, verification reminders. Read the tooltip/banner text.
⚠ (red text)	Active error or blocking condition; details appear next to the affected step.
✓ (green)	Step completed for the active restoration.
⏳ / ✓ / ⚠	AI segmentation job status: running / results ready / failed.
FDI numbers (e.g. 11, 36)	Tooth notation assigned to each tooth by auto-tagging or manually.
👁	Toggle object visibility in the 3D view.
🗑	Delete object (confirmation required where destructive).
① ② ③ …	Numbered callouts in this manual's annotated screenshots.

⚠️ **Caution** A symbol shown in the running software always takes precedence over its description here. As a demonstration build (chapter 1.1), DWOS Web carries no medical-device symbols (CE mark, UDI, manufacturer or "consult instructions for use" markings).

Mouse and keyboard navigation in the CAD workstation: **left-drag** orbit · **middle-drag** or **wheel-press** pan · **wheel** zoom · **Esc** cancel the current tool / close dialogs.